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Ex.No. 14: 5G Throughput vs SNR with Modulation Labels

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AIM: To simulate the performance of a **5G wireless communication system** using MATLAB by analyzing the relationship between **Throughput (Mbps)**, **Signal-to-Noise Ratio (SNR) (dB)**, and **different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM)**.

Objective:

1. To evaluate the **Throughput (Mbps)** of a 5G communication system under different Signal-to-Noise Ratio (SNR) conditions.
2. To analyze the effect of **Signal-to-Noise Ratio (SNR) (dB)** on the performance and reliability of the 5G communication system.
3. To compare the performance of different **Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM)** and determine their suitability for varying SNR conditions.

Procedure

1. Define the simulation parameters such as SNR range, bandwidth, modulation schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM), and transmission settings.
2. Generate random binary data and modulate it using the selected modulation schemes.
3. Simulate the transmission of the modulated signals through an Additive White Gaussian Noise (AWGN) channel with different SNR values.
4. Calculate the **Throughput (Mbps)** achieved for each modulation scheme at different SNR levels.
5. Analyze the impact of **Signal-to-Noise Ratio (SNR)** on communication performance and throughput.
6. Compare the performance of QPSK, 16-QAM, 64-QAM, and 256-QAM under identical channel conditions.
7. Plot MATLAB graphs for **Throughput vs SNR** and **SNR vs Modulation Scheme** to visualize the system performance.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')
grid on
```



The throughput of the 5G system increases as the SNR increases due to improved signal quality. Higher SNR enables efficient data transmission, resulting in greater throughput and better communication performance

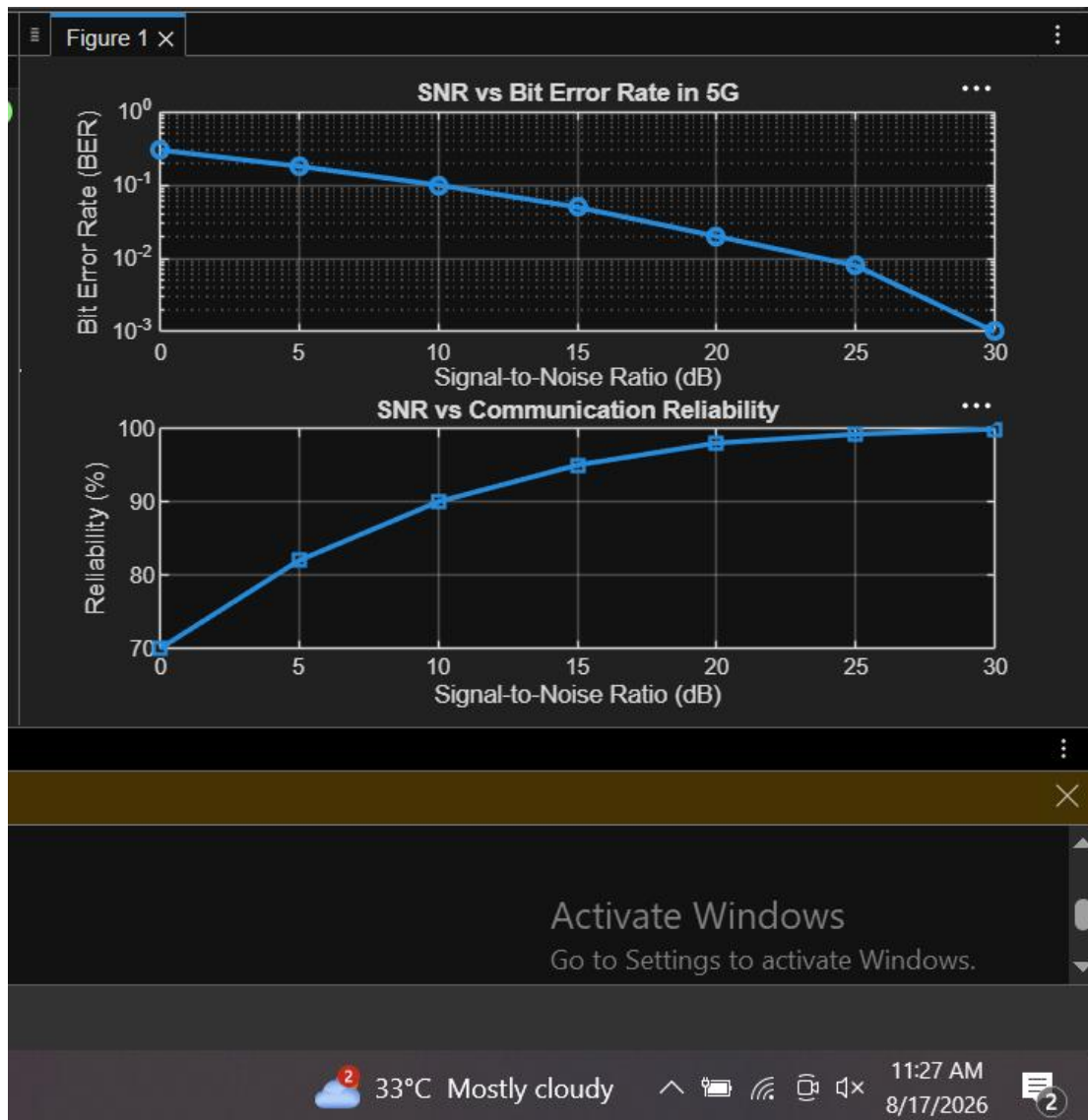
Objective 2 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
% Reliability (%)
Reliability = (1-BER)*100;
```

```

disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
semilogy(SNR,BER,'-o','LineWidth',2)
title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,Reliability,'-s','LineWidth',2)
title('SNR vs Communication Reliability')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Reliability (%)')
grid on

```



As the **Signal-to-Noise Ratio (SNR)** increases, the **Bit Error Rate (BER)** decreases, resulting in more reliable communication. Higher SNR improves the quality of the received signal, thereby enhancing the overall performance of the 5G communication system.

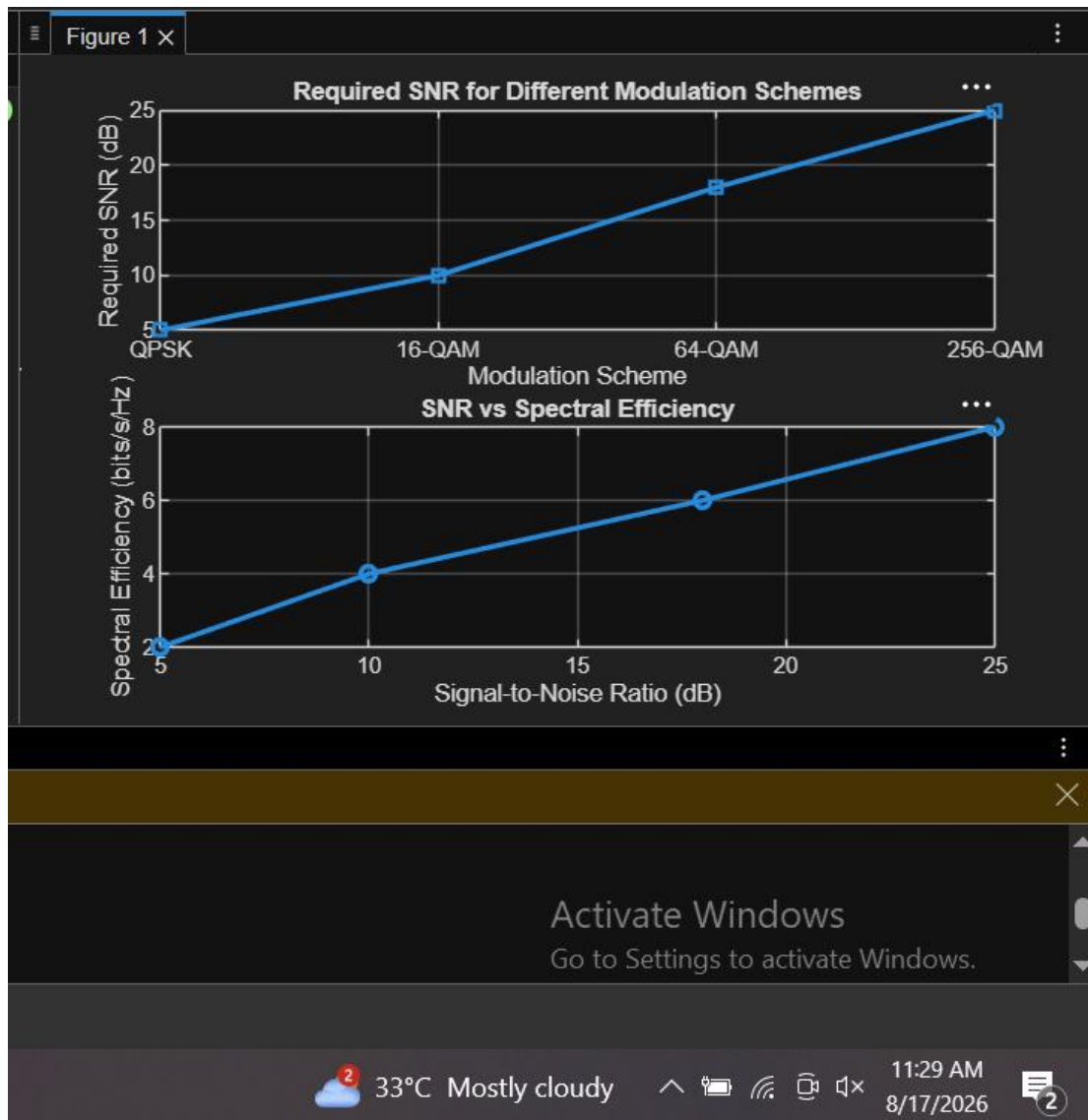
Objective 3 Matlab Code and Output:

```
clc;
clear;
close all;
% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
% Required SNR (dB)
SNR = [5 10 18 25];
```

```

% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'})))
figure
% ----- Graph 1 -----
subplot(2,1,2)plot(SNR,Efficiency,'-o','LineWidth',2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
% ----- Graph 2 -----
subplot(2,1,1)
plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on

```



Result:

1. The **Throughput (Mbps)** of the 5G communication system increased with increasing **Signal-to-Noise Ratio (SNR)**, demonstrating improved data transmission performance.
2. The simulation showed that higher **SNR (dB)** reduced transmission errors and enhanced the reliability and quality of 5G communication.
3. The comparison of **QPSK, 16-QAM, 64-QAM, and 256-QAM** revealed that higher order modulation schemes achieved greater throughput and spectral efficiency but required higher SNR for reliable communication.